

A Packing of 6 Four-Spaces in $\mathbb{R}^{13}$

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Abstract

We give 6 4-dimensional subspaces of $\mathbb{R}^{13}$ whose smallest squared chordal distance is at least 3.322896. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 3.230536, obtained by deleting 1 of the 7 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 6 subspaces of dimension 4 in $\mathbb{R}^{13}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 3.322896 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 13$, $n = 4$. The cell $N = 6$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 3.230536, obtained by deleting 1 of the 7 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(13 - 4)/13 \cdot 6/5 = 3.323077$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = Y G^{-1} Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m13n4N6.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
103840510352333969267388060074070880822417780673211233568430327552784407
625333995087420796609351892892877272787156907372403826613232808789355611
464479761236345513815236923863061586143956693301
```

denominator:

```
312499999999595169093970323923682639065738542077414655380427566046972312
418965614762716475706969202270046315480388688718890010768800120247194141
88776020239074913959745915093511299362224196722
```

B Coordinates

6 blocks of 13 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m13n4N6.txt`.

```
251521316579 -106810332313 408699660865 54558499067292614359242 -71485158616 297396669135 -14553883973718894933386 -606218710284 -165273451907 51901588589
-218178299268 148973881631 124971651783 -834978978733325365522 -19206751672 -659344309729 26606764226-97968171280 -88927151134 110499123286 -63525963954
-126390312716 -108367837086 123884520049 2541904682537989801384 -20815815264 144210059556 -13959040650-266514750760 155890538753 376401989566 506151135843
-92737263007 629108844995 -162004044035 39259952817386257073782 93540435281 205151825783 -108369006397-2384960291 161080377690 48387573722 -507828259610
-456520199733 103841352628 163096603035 13379262177116396914633 -8052935370 -187948280093 630819758687120121525524 102029739055 -300185917006 236610567792
-384863682609 -127162338061 -523979212949 2088321865961152030 -294915764854 135776585197 19414059580461112152340 -245798460412 266095220065 42186854472
25078602927 14822682740 54878042054 341620910569 -444030680665 -87921497857 -510364953274 -395308847555627136717 -243143602194 353666722423 -333787687646
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442352929593 32436873140 -176603651444 -29154580866382761356780 -651181345654 -79638421540 11512845433361870778395 -231440523489 83938973143 -308052951353
341128938264 260001525249 -566791838005 20032596445713603590133 491126228593 30450584210 224212706743 -30886392513 -11532233785 -232276237761 154365707048
-131962744353 -505522473262 -258300738247 5165083463016093750 -23525790759 242645940473 602591130409 -194789344393 -4184346614 403833699995 -151078083036
226076681709 268305864030 221023638000 -54878359585-35914588372 184465752180 -110763487576 -2174395035302046195103 514777711209 357963614779 11076144090
-323987338336 359140760699 57326882404 -40727936743874852371 170292695912 20492836344 209872211621 287297445245 -329086020344 314548216462 404734604467
324631254238 -49333206193 -83381410460 33626312528 71621512225 -127392211330 399616353852 73555112555 381289534153 -193355475487 227674280626 28993810777
378712006065 -295610208260 153152805112 -46055628907269391689 134962384827 -135793651631 158152624259-249305485662 450013685297 218023658263 302025893421
32269015097 -374257558899 590970565342 -54110188063-56068378522 429447866971 111326753543 -33136097222-216772565486 -536584480640 -305248120242 308648436456
-214298830231 426034561802 194682044525 -3795298338200418260037 -883563389 277356936638 -471040144991-132981481273 -88452517718 16901367143 232067421854
-271533461070 285629468940 -3800073656 163822732129-110638357059 -86849432040 271255240543 410102251306178241376264 -257875665252 642170601781 101933956999
140556966691 -144126407180 146640389067 -1341104945712573215446 -318593869048 213981353125 -1752261605363140852367 -64496012186 91695221498 -434746359593
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153470584809 -148782354282 -582439446013 -164405512226528435869 729142178894 32079670710 -2032220539653620527795 -94715760645 253293564705 388383435292
-376514261822 -208018718644 181744600994 21484469786321525360334 38561432391 48839876828 294115571464 -46258871854 -286981461772 25411873363 351354578319
-211046325314 -320569668122 -287154530185 -1665889253252873139 203326557652 412613019357 514204056862-75265990825 150901246523 -370775054881 4747798386
229723454268 79023182486 -235863684720 257042111564-316852204856 493665687410 230084835170 -2357481810613433057691 460741751907 -111171422734 467458413892
143404525863 -359980580519 36621471997 5514430251818770206726 1971137389 615822225706 -144556861398 -29289978497 241151009376 410473958896 -139741693194
-216909449038 -236834290938 -159692566843 -35187078485773996781 52477548492 75508265392 -25019547910792296884116 -93467515399 53478644183 -136559084877
```